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Direct trajectory optimization framework for vertical takeoff and vertical landing reusable rockets: case study of two-stage rockets

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ABSTRACT

This study presents a direct trajectory optimization framework for a vertical takeoff and vertical landing (VTVL) reusable rocket with a two-stage structure for a launch mission to simultaneously deliver a payload to the desired orbit and recover the rocket. A two-phase trajectory optimization problem of VTVL reusable rockets is established, with the main features of the flight process. The finite-element collocation approach with Radau collocation is utilized to transcribe the established problem into a nonlinear programming (NLP) problem solved by Interior Point OPTimizer (IPOPT). Novel internal-external-growth initialization strategies are designed to enhance the convergence of solving the NLP problem. The proposed direct trajectory optimization framework is applied to two representative scenarios of VTVL reusable rockets: return to a launch site and return to a drone ship. Simulation results illustrate that the proposed trajectory optimization framework has the adaptability to deal effectively with the complex missions of VTVL reusable rockets.

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Trajectory optimization; VTVL reusable rocket; finite-element collocation approach; IPOPT; internal-external-growth strategies

1. Introduction

Rockets are typically destroyed on their maiden voyage. Making rockets practically reusable to significantly reduce the cost of space exploration has gained worldwide attention. Private aerospace companies, such as SpaceX and Blue Origin, have made remarkable progress in the technology of reusable rockets. The vertical takeoff and vertical landing (VTVL) scheme has become a promising recovery technique. Rockets can make an upright landing and will be refuelled for another trip, which is setting the stage for a new era in spaceflight (Liu 2017). Reusable rockets have been conceived and discussed since the 1990s (Cook 1995; Gallaher, Coughlin, and Krupp 1996) and were practically demonstrated to be achievable in recent years. The trajectory design for the entire flight process is of great importance to recover rockets to ensure that the upper stage payload can be delivered to the desired orbit and the lower stage can land at a specified site. The entire flight process of reusable rockets generally includes several flight phases, and the upper and lower stages should be considered simultaneously to maximize flight performance.

The trajectory design for the entire flight process of VTVL reusable rockets primarily involves two motion types: powered ascent and descent. Various studies have investigated powered ascent and descent. The Saturn/Apollo ascent guidance was known as Iterative Guidance Mode, based on

calculus of variations (Chandler and Smith 1967). Cohen and Brown (1973) developed a guidance algorithm *in vacuo* which solved the nonlinear two-point boundary value problem and demonstrated its effectiveness for performing both ascent and orbital powered manoeuvres with a minimum of simplifying assumptions. McHenry *et al.* (1979) proposed a vector version of Iterative Guidance Mode named Powered Explicit Guidance for space shuttle vacuum ascent and orbit manoeuvring, which used linear vector steering combined with numerical integration to accurately obtain gravity effects over long burn arcs. Gath and Calise (2001) described improvements made to a hybrid analytical/numerical algorithm for optimizing launch vehicle ascent trajectories. Zhang and Lu (2008) presented a fixed-point formulation and the associated algorithms for rapidly generating optimal ascent trajectories of launch vehicles through the atmosphere. Lu *et al.* (2008) provided detailed development of an analytical multiple-shooting method for rapid and reliable generation of the optimal exoatmospheric ascent trajectory of a launch vehicle. Ma *et al.* (2016a) proposed a unified trajectory optimization framework for lunar ascent based on direct methods; moreover, a homotopy-based backtracking initial value strategy was designed to enhance the convergence of the proposed trajectory optimization framework.

For the powered descent problem, Meditch (1964) presented a closed-form solution of the lunar soft-landing problem for a one-dimensional case. Topcu, Casoliva, and Mease (2005) developed first order necessary conditions for a fuel-efficient Mars landing; the optimal thrust profile included a maximum–minimum–maximum profile. Direct collocation and direct multiple shooting methods were used to obtain numerical solutions of a constrained non-convex parameter optimization problem converted from the original guidance problem. Sostaric and Rea (2005) developed a numerical solution to the pinpoint-landing guidance problem by Legendre pseudospectral methods. Najson and Mease (2005) developed an approximate analytical solution for soft landing by solving a related optimal control problem that did not use a minimum fuel cost functional. Açıkmeşe and colleagues (Açıkmeşe and Ploen 2007; Açıkmeşe and Blackmore 2011; Açıkmeşe *et al.* 2013) developed powered descent guidance using convex optimization. Lossless convexification of non-convex control bound and pointing constraints of the soft-landing optimal control problem was proposed. The resulting convex optimal problem was solved by efficient second order cone programming solvers with deterministic convergence properties. Ma *et al.* (2016b, 2018) optimized the trajectory of the lunar soft-landing process via finite-element collocation approaches, and a Hamiltonian-based adaptive mesh refinement strategy was proposed to catch the breakpoints of the thrust profile. Liu (2017) optimized the fuel-optimal rocket landing problem in the vertical plane with the angle of attack, thrust magnitude and thrust direction of the rocket as control inputs based on convex optimization. Ma *et al.* (2017) proposed a trajectory optimization framework based on optimal sensitivity for planetary multi-point soft landings.

Numerical methods for trajectory optimization problems are mainly divided into two general categories: indirect methods and direct methods (Betts 1998). Indirect methods have several disadvantages, including the need to derive the Hamiltonian boundary value problem analytically, a small region of convergence of solving the Hamiltonian boundary value problem using a root-finding algorithm, a non-intuitive initial guess for the costate, and *a priori* knowledge of the constrained and unconstrained arcs if path constraints are present (Benson *et al.* 2006; Betts 1998). Moreover, most trajectory optimization problems do not have an analytical solution. The key to trajectory optimization strategies is the need to determine optimal trajectories for complex system models with efficient and reliable nonlinear programming (NLP) methods (Chen, Shao, and Biegler 2014); hence, direct methods are preferred to solve complex trajectory optimization problems. Direct methods for trajectory optimization do not require explicit consideration of the necessary conditions (adjoint equations, transversality conditions and maximum principle) (Betts 1998). Pseudospectral methods (Benson *et al.* 2006; Garg *et al.* 2010), which are a class of direct methods where the trajectory optimization problem is transcribed into an NLP problem, have recently become increasingly popular and are widely used to obtain the numerical solution of trajectory optimization problems. The state and control variables are parameterized using global polynomials at collocation nodes derived

from a Gaussian quadrature. For problems with smooth solutions, the application of global polynomials associated with Gaussian quadrature collocation points provides accurate approximations and exponential convergence (Garg *et al.* 2010). However, for problems whose solutions are non-smooth or not well approximated by global polynomials, finite-element collocation approaches (Biegler 2007; Kameswaran and Biegler 2006, 2007) are preferred, where the time interval is partitioned into subintervals and polynomials are used to approximate the state and control profiles over each subinterval.

In this study, a direct trajectory optimization framework is proposed to handle complex missions of VTVL reusable rockets. The reusable rocket is assumed to be a two-stage structure. A two-phase trajectory optimization problem is formulated to describe the problem of VTVL reusable rockets, and the trajectory optimization problem consists of the following: (1) differential equations that describe the kinematics and dynamics of rockets; (2) algebraic equalities/inequalities that define the boundary, path and interior-point constraints of flight process; and (3) a specified objective function. Analytical solutions generally cannot be obtained because of the complexity and unification of the formulated trajectory optimization problem. Thus, the direct method is selected to solve the problem. For the fuel-optimal powered descent of the first stage of a reusable rocket with the bounded magnitude of thrust, the optimal thrust profile typically has a ‘bang–bang’ profile (Lee 2011). Breakpoints exist in the optimal thrust profile, *i.e.* non-smooth solutions; hence, the finite-element collocation approach is selected to transcribe the trajectory optimization problem of VTVL reusable rockets into an NLP problem solved by Interior Point OPTimizer (IPOPT) (Wächter and Biegler 2006). The NLP solver based on Newton’s method is often sensitive to the guess of initial value; thus, a good initial value strategy is beneficial for successful solving of the resulting NLP problem. Internal–external-growth initialization strategies are designed to simultaneously consider discretization of the finite-element collocation approach and the character of IPOPT and enhance the convergence of solving the formulated two-phase trajectory optimization problem of VTVL reusable rockets with complex constraints.

The rest of this article is organized into the following sections. Section 2 formulates the two-phase trajectory optimization problem of VTVL reusable rockets. Section 3 introduces the direct trajectory optimization framework combined with the finite-element collocation approach and internal–external-growth strategies to solve the formulated trajectory optimization problem. Section 4 presents the two designed representative scenarios, simulation results and discussion. Section 5 concludes the study.

2. Problem formulation

2.1. Point-mass dynamic model

The reusable rocket in this study is a two-stage rocket. The point-mass dynamic model of the rocket is expressed in an Earth-centred inertial frame, with several simplifying assumptions (Benson 2005):

- (1) The thrust magnitude does not depend on the atmospheric pressure, under the assumption that the thrust from each engine is the vacuum thrust.
- (2) The reference area and drag coefficient are assumed to be constant for the entire trajectory and not to depend on the Mach number or angle of attack.
- (3) No component of lift exists. The drag is assumed always to be in the opposite direction of the velocity and not to depend on the rocket orientation.
- (4) The Earth is assumed to be a sphere, which facilitates determining the position of the launch site and the altitude above the Earth.
- (5) The spherical Earth is assumed to satisfy the point-mass gravity model.

Under these simplifying assumptions, the dynamic model is given as follows:

$$\begin{aligned}\dot{r} &= v \\ \dot{v} &= -\frac{\mu}{\|r\|^3}r + \frac{T}{m} + \frac{D}{m} \\ \dot{m} &= -\frac{\|T\|}{g_0 I_{sp}}\end{aligned}\quad (1)$$

where $r = (x, y, z)$ is the position vector, $v = (v_x, v_y, v_z)$ is the velocity vector, $T = (T_x, T_y, T_z)$ is the thrust vector, and $D = (D_x, D_y, D_z)$ is the drag force vector. μ is the gravitational parameter, g_0 is the gravity acceleration at sea level, and I_{sp} is the specific impulse of the engine. The drag force vector is defined as follows:

$$D = -\frac{1}{2}C_D A_{ref} \rho \|v\| v \quad (2)$$

where C_D is the drag coefficient, A_{ref} is the reference area, and ρ is the atmospheric density. The atmospheric density is defined as follows:

$$\rho = \rho_0 e^{-h/h_0} \quad (3)$$

where ρ_0 is the atmospheric density at sea level, $h = \|r\| - R_e$ is the altitude, R_e is the equatorial radius of the Earth, and h_0 is the density scale height.

After stage separation of the first and second stages, the second stage will continue to deliver the payload to the desired orbit, while the first stage will return to land at a specified site. For the convenience of establishing the trajectory optimization problem, the first and second stages are modelled separately. (r_1, v_1, m_1) and T_1 denote the state and control variables of the first stage, respectively; (r_2, v_2, m_2) and T_2 denote the state and control variables of the second stage, respectively. The motion of the two stages satisfies the above point-mass dynamic model.

2.2. Constraints

The entire flight process is divided into two phases by the stage separation of the reusable rocket. Here, $(r_1^{(1)}, v_1^{(1)}, m_1^{(1)}, T_1^{(1)})$ and $(r_1^{(2)}, v_1^{(2)}, m_1^{(2)}, T_1^{(2)})$ represent the state and control variables of the first stage in the first and second phases, respectively. $(r_2^{(1)}, v_2^{(1)}, m_2^{(1)}, T_2^{(1)})$ and $(r_2^{(2)}, v_2^{(2)}, m_2^{(2)}, T_2^{(2)})$ represent the state and control variables of the second stage in the first and second phases, respectively.

2.2.1. Initial constraints

The two-stage rocket lifts off the ground at time t_0 ; therefore, the initial constraints are

$$\begin{aligned}r_1^{(1)} &= r_2^{(1)}(t_0) = r_0 \\ v_1^{(1)}(t_0) &= v_2^{(1)}(t_0) = 0 \\ m_1^{(1)}(t_0) &= m_{10}, \quad m_2^{(1)}(t_0) = m_{20}\end{aligned}\quad (4)$$

where m_{10} and m_{20} are the initial masses of the first and second stages, respectively, and r_0 refers to the position of launch site.

2.2.2. Terminal constraints

The first stage of the rocket returns to the Earth and finally lands at a specified site at time t_{f1} after stage separation; thus, the terminal constraint of the first stage is given as follows:

$$r_1^{(2)}(t_{f1}) = r_{f1}$$

$$v_1^{(2)}(t_{f1}) = 0 \quad (5)$$

The second stage of the rocket finally delivers the payload to the desired orbit at time t_{f2} , and the terminal position and velocity of the second stage can be transformed into orbital elements. The terminal constraint of the second stage is hence defined as follows:

$$\begin{aligned} a_2^{(2)}(t_{f2}) &= a_{f2} \\ e_2^{(2)}(t_{f2}) &= e_{f2} \\ i_2^{(2)}(t_{f2}) &= i_{f2} \\ \Omega_2^{(2)}(t_{f2}) &= \Omega_{f2} \\ \omega_2^{(2)}(t_{f2}) &= \omega_{f2} \end{aligned} \quad (6)$$

where the orbital elements a, e, i, Ω, ω represent the semi-major axis, eccentricity, inclination, right ascension of the ascending node and argument of perigee, respectively. The true anomaly is left undefined because the exact location within the orbit is not constrained.

2.2.3. Interior-point constraints

The two phases of the entire flight process need to be linked by the interior-point constraint as follows:

$$r_i^{(1)}(t_{sp}) = r_i^{(2)}(t_{sp}), v_i^{(1)}(t_{sp}) = v_i^{(2)}(t_{sp}), m_i^{(1)}(t_{sp}) = m_i^{(2)}(t_{sp}) \quad i = 1, 2 \quad (7)$$

where t_{sp} is the time of stage separation.

2.2.4. Path constraints

The rocket altitude is always above the Earth surface; thus, a state constraint is imposed as follows:

$$\|r_i(t)\| \geq R_e \quad i = 1, 2 \quad (8)$$

The glide slope constraint (Açıkmeşe, Carson, and Blackmore 2013; Szmuk, Eren, and Açıkmeşe 2017) is imposed to ensure that the first stage of the rocket stays at a safe distance from the ground until it reaches the specified site, and is described as follows:

$$(r_1^{(2)}(t) - r_{f1})^T r_{f1} \geq \|(r_1^{(2)}(t) - r_{f1})\| \|r_{f1}\| \cos \theta_{gs} \quad (9)$$

where θ_{gs} is the glide slope angle.

Fuel consumption cannot be more than the limitative fuel mass; therefore, this path constraint is given by

$$m_i(t) \geq m_{i0} - m_{ifuel} \quad i = 1, 2 \quad (10)$$

where m_{1fuel} and m_{2fuel} are the limitative fuel masses of the first and second stages, respectively.

In this study, the first stage incorporates nine engines and the second stage is powered by a single engine. The first stage is in the full-thrust performance before stage separation and the second stage is also in the full-thrust performance after stage separation. After stage separation, the first stage utilizes one engine in the throttleable performance to return. Therefore, the path constraint for thrust magnitude is given as follows:

$$\begin{aligned} \|T_1^{(1)}\| &= 9T_{\max}, \|T_2^{(1)}\| = 0 \\ T_{\min} &\leq \|T_1^{(2)}\| \leq T_{\max}, \|T_2^{(2)}\| = T_{\max} \end{aligned} \quad (11)$$

where $T_{\min}$ and $T_{\max}$ are the minimum thrust and maximum thrust of one engine, respectively.

The dynamic pressure and visual acceleration of the rocket in the flight process are constrained as follows:

$$\begin{aligned} \frac{1}{2} \rho \|v_i\|^2 &\leq q_{\max} \\ \left\| \frac{T_i + D_i}{m_i} \right\| &\leq acc_{\max} \quad i = 1, 2 \end{aligned} \quad (12)$$

where $q_{\max}$ and $acc_{\max}$ are the maximum dynamic pressure and maximum visual acceleration, respectively.

2.3. Objective function

The objective in this study is to maximize the remaining fuel in the second stage of the rocket (*i.e.* to maximize the potential payload); thus, the objective function is to minimize

$$J = -m_2^{(2)}(t_{f2}) \quad (13)$$

This objective function, along with the dynamic model, boundary, path and interior-point constraints, makes up the trajectory optimization problem of the VTVL reusable rocket, which involves two free terminal times for the first and second stages and the free time for stage separation.

3. Direct trajectory optimization framework for VTVL reusable rockets

A direct trajectory optimization framework for the two-phase trajectory optimization problem of VTVL reusable rockets established in Section 2 is presented in this section. The finite-element collocation approach is utilized to discretize the original trajectory optimization problem into an NLP problem solved by IPOPT. A series of complex constraints, such as multi-phase, path and boundary constraints, results in a small feasible region of the discretized trajectory optimization problem. The NLP solver based on Newton's method often fails to converge; thus, an advanced initialization strategy is proposed to help the solver to overcome the convergence difficulty.

3.1. Finite-element collocation approach

The finite-element collocation approach fully discretizes the state and control variables, thereby leading to large-scale NLP problems. Karush–Kuhn–Tucker conditions of the resulting NLP problem are consistent with the discretized Euler–Lagrange equations of the original trajectory optimization problem. Therefore, the optimal solution of this NLP problem is an approximate optimal solution of the original trajectory optimization problem (Kameswaran and Biegler 2007). The authors prefer Radau collocation points because they allow constraints to be set at the end of each element and to stabilize the system efficiently if high-index differential algebraic equations (DAEs) are present. Without loss of generality, the following general trajectory optimization problem is considered (Biegler 2007; Kameswaran and Biegler 2006):

$$\begin{aligned} \min \quad & \Phi(\mathbf{Z}(t_f)) \\ \text{s.t.} \quad & \dot{\mathbf{Z}} = \mathbf{f}(\mathbf{Z}(t), \mathbf{U}(t)), \mathbf{Z}(t_0) = \mathbf{Z}_0 \\ & \mathbf{g}(\mathbf{Z}(t), \mathbf{U}(t)) = 0 \\ & \mathbf{U}_L \leq \mathbf{U}(t) \leq \mathbf{U}_U \\ & \psi(\mathbf{Z}(t_f)) \leq 0 \end{aligned} \quad (14)$$

where $\mathbf{Z}(t)$ and $\mathbf{U}(t)$ denote the state and control profiles, respectively. For the formulated problem in Section 2, $\mathbf{Z}$ and $\mathbf{U}$ refer to state variables (r, v, m) and control variables T , respectively. t_f is the final time, $\Phi(\mathbf{Z}(t_f))$ is an objective function, g refers to the algebraic equation constraints, and ψ indicates the terminal constraints. The DAE model is assumed to be index-1 and given in semi-explicit form. The entire time horizon is divided into NE finite elements, which satisfies

$$t_0 < t_1 < t_2 < \cdots < t_{NE-1} < t_{NE} = t_f \quad (15)$$

$K + 1$ Radau interpolation points are selected in finite element i . The state and control profiles in a specified finite element i are approximated by the Lagrange polynomial as follows:

$$\begin{aligned} \mathbf{Z}^K(t) &= \sum_{j=0}^K L_j(\tau) \mathbf{Z}_{ij}, \quad \mathbf{U}^K(t) = \sum_{j=1}^K \bar{L}_j(\tau) \mathbf{U}_{ij} \\ L_j(\tau) &= \prod_{k=0, \neq j}^K \frac{(\tau - \tau_k)}{(\tau_j - \tau_k)}, \quad \bar{L}_j(\tau) = \prod_{k=1, \neq j}^K \frac{(\tau - \tau_k)}{(\tau_j - \tau_k)} \\ t &\in [t_{i-1}, t_i], \quad t = t_{i-1} + h_i \tau, \quad \tau \in [0, 1] \end{aligned} \quad (16)$$

where h_i refers to the length of the finite element i , and $0 < \tau_j \leq 1, j = 1, \dots, K$ are the shifted Radau points. This polynomial representation has the following properties:

$$\begin{aligned} \mathbf{Z}_{ij} &= \mathbf{Z}^K(t_{ij}), \quad \mathbf{U}_{ij} = \mathbf{U}^K(t_{ij}) \\ t_{ij} &= t_{i-1} + h_i \tau_j \end{aligned} \quad (17)$$

The continuity of the state profile at the finite-element boundaries is enforced by the following expression:

$$\begin{aligned} \mathbf{Z}_{i+1,0} &= \sum_{j=0}^K L_j(1) \mathbf{Z}_{ij}, \quad i = 1, \dots, NE - 1 \\ \mathbf{Z}(t_f) &= \sum_{j=0}^K L_j(1) \mathbf{Z}_{Nj}, \quad \mathbf{Z}_{1,0} = \mathbf{Z}_0 \end{aligned} \quad (18)$$

Equations (16)–(18) are substituted into Problem (14), and collocation equations can be derived as follows:

$$\begin{aligned} \sum_{j=0}^K \frac{dL_j(\tau_k)}{d\tau} \mathbf{Z}_{ij} - h_i f(\mathbf{Z}_{ik}, \mathbf{U}_{ik}) &= 0 \\ \mathbf{Z}_{i+1,0} &= \sum_{j=0}^K L_j(1) \mathbf{Z}_{ij} \\ g(\mathbf{Z}_{ik}, \mathbf{U}_{ik}) &= 0, \quad i = 1, \dots, NE, k = 1, \dots, K \end{aligned} \quad (19)$$

The trajectory optimization problem is discretized into an NLP formulation with fixed finite-element length h_i , as follows:

$$\min \Phi(\mathbf{Z}(t_f))$$

$$\begin{aligned}
& \text{s.t. } \sum_{j=0}^K \frac{dL_j(\tau_k)}{d\tau} \mathbf{Z}_{ij} - h_{if}(\mathbf{Z}_{ik}, \mathbf{U}_{ik}) = 0 \\
& g(\mathbf{Z}_{ik}, \mathbf{U}_{ik}) = 0 \\
& \mathbf{U}_L \leq \mathbf{U}_{ik} \leq \mathbf{U}_U, \quad i = 1, \dots, NE, k = 1, \dots, K \\
& \psi(\mathbf{Z}(t_f)) \leq 0 \\
& \mathbf{Z}_{i+1,0} = \sum_{j=0}^K L_j(1) \mathbf{Z}_{ij}, \quad i = 1, \dots, NE - 1 \\
& \mathbf{Z}(t_f) = \sum_{j=0}^K L_j(1) \mathbf{Z}_{Nj}, \mathbf{Z}_{1,0} = \mathbf{Z}_0
\end{aligned} \tag{20}$$

3.2. Internal-external-growth strategies

The resulting large-scale NLP problem obtained by the finite-element collocation approach, which discretizes the state and control variables of the original trajectory optimization problem, will be solved by the NLP solver IPOPT, and the two-phase trajectory optimization problem of VTVL reusable rockets involves complex constraints. The resulting large-scale NLP problem with complex constraints is always sensitive to the guess of initial value. In this subsection, advanced initialization strategies are presented to generate a good guess of initial value for the resulting NLP problem, which helps the solving process to converge successfully to an optimal solution.

The three main ideas of the presented internal-external-growth strategies are as follows: (1) from a small-scale problem to a large-scale problem; (2) from a problem with simple constraints to a problem with complex constraints; and (3) associating with the algorithm of IPOPT. Therefore, several subproblems of the original trajectory optimization problem will be solved to obtain a good guess of initial value. For the general trajectory optimization problem in Equation (14), the resulting NLP problem is Equation (20). The number of variables and equalities/inequalities of the resulting NLP problem depends on the number of finite elements and the number of collocation points in each finite element.

The basic algorithm of IPOPT is briefly introduced. IPOPT is utilized to solve the NLP problem as follows (Wächter and Biegler 2006):

$$\begin{aligned}
& \min_{\mathbf{X} \in \mathbb{R}^n} f(\mathbf{X}) \\
& \text{s.t. } c(\mathbf{X}) = 0 \\
& \mathbf{X} \geq 0
\end{aligned} \tag{21}$$

where $\mathbf{X}$ is the vector which consists of all the discretized variables in Equation (20), and n is the total number of all variables. IPOPT implements an interior-point filter line-search method that aims to find a local optimal solution. The interior-point method transforms the problem in Equation (21) into a sequence of barrier problems as

$$\begin{aligned}
& \min_{\mathbf{X} \in \mathbb{R}^n} f(\mathbf{X}) - \epsilon \sum_{i=1}^n \ln(\mathbf{X}^{(i)}) \\
& \text{s.t. } c(\mathbf{X}) = 0
\end{aligned} \tag{22}$$

with a decreasing sequence of barrier parameter ϵ converging to zero. Equivalently, the problem in Equation (22) can be interpreted as utilizing a homotopy method to the primal–dual equations as

$$\begin{aligned}\nabla_X f(\mathbf{X}) + \nabla_X c(\mathbf{X})\lambda - \gamma &= 0 \\ c(\mathbf{X}) &= 0 \\ \mathbf{X}^{(i)}\gamma^{(i)} &= \epsilon\end{aligned}\quad (23)$$

where $\lambda \in R^m$, $\gamma \in R^n$ are the Lagrangian multipliers for the equality and bound constraints, respectively. The optimality error for the barrier problem is defined as

$$E_\epsilon(\mathbf{X}, \lambda, \gamma) = \max \left\{ \frac{\|\nabla_X f(\mathbf{X}) + \nabla_X c(\mathbf{X})\lambda - \gamma\|_\infty}{s_d}, \|c(\mathbf{X})\|_\infty, \frac{\|A\mathbf{B}e - \mu e\|_\infty}{s_c} \right\} \quad (24)$$

where the notation $A = \text{diag}(\mathbf{X})$; the notation $B = \text{diag}(\gamma)$; e stands for the vector of all ones; and s_d , s_c are scaling parameters.

A damped Newton's method is utilized in the primal–dual equations (Equation (23)) to solve the barrier problem (Equation (22)) for a fixed-barrier parameter ϵ_j . The detailed method for obtaining the solution of the barrier problem is given by Wächter and Biegler (2006). The strategy for updating the barrier parameter is

$$\epsilon_{j+1} = \max \left\{ \frac{\vartheta_{tol}}{10}, \min(k_\epsilon \epsilon_j, \epsilon_j^{\theta_\epsilon}) \right\} \quad (25)$$

with constants $k_\epsilon \in (0, 1)$, $\theta_\epsilon \in (1, 2)$ and the given desired tolerance ϑ_{tol} . In this way, the barrier parameter is decreased as a superlinear rate. The optimal solution $(\mathbf{X}^*, \lambda^*, \gamma^*)$ can be obtained, which satisfies

$$E_0(\mathbf{X}^*, \lambda^*, \gamma^*) \leq \vartheta_{tol} \quad (26)$$

The optimal solution involves the values of primal variables $\mathbf{X}^*$ and dual variables (λ^*, γ^*) .

One of the main difficulties in solving the large-scale NLP problem formulated as Equation (21) arises from the fact that full-space variable information is required to generate the guess of initial value of primal variables x and the values of dual variables (λ, γ) ; therefore, initialization strategies are necessary for addressing the trajectory optimization problem using direct methods. The proposed internal–external-growth strategies are divided into two parts: external-growth and internal-growth strategies. In the external-growth part, the low-density discretized solution of the trajectory optimization problem without path constraints, such as dynamic pressure, visual acceleration and glide slope constraints, is derived. The low-density discretized solution is utilized as the guess of initial value to solve the entire trajectory optimization problem of the VTVL reusable rocket. After the optimal low-density discretized solution has been derived, the internal-growth part is activated. After that, in the external-growth part, the trajectory optimization problem is discretized by high-density collocation points; thus, the scale of the discretized problem is enlarged. In the internal-growth part, not only the optimal low-density discretized solution, but also the corresponding multipliers are utilized to generate the guess of initial value to facilitate the convergence of solving the high-density discretized problem.

3.3. Direct trajectory optimization framework

In this subsection, the overall direct trajectory optimization framework illustrated in Figure 1 for VTVL reusable rockets is concluded as follows:

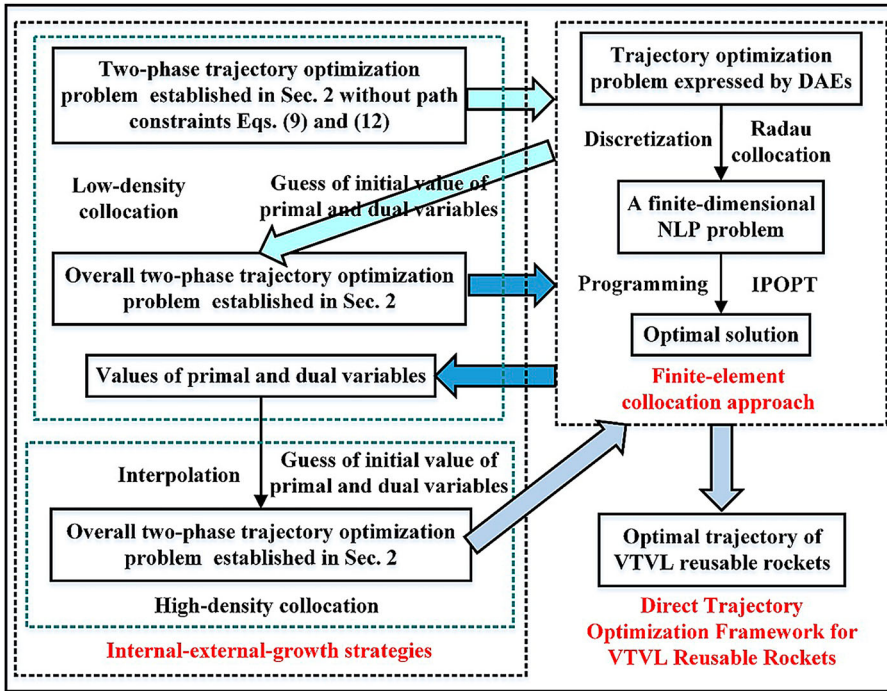


Figure 1. Direct trajectory optimization framework for vertical takeoff and vertical landing (VTVL) reusable rockets. DAEs = differential algebraic equations; NLP = nonlinear programming; IPOPT = Interior Point OPTimizer.

- Step 1: The trajectory optimization problem of VTVL reusable rockets without dynamic pressure, visual acceleration and glide slope constraints is discretized using the finite-element collocation approach with low-density collocation points, and the resulting NLP problem is solved by IPOPT.
- Step 2: The entire trajectory optimization problem of VTVL reusable rockets is transcribed using the same low-density collocation points in Step 1 into an NLP problem solved by IPOPT with guess of initial value of primal and dual variables given by the solution in Step 1.
- Step 3: The optimal solution of the trajectory optimization problem obtained in Step 2, *i.e.* the values of primal and dual variables of the resulting NLP problem, is stored.
- Step 4: The entire trajectory optimization problem is discretized by the finite-element collocation approach with high-density collocation points, then the resulting NLP problem is solved based on the guess of initial value given by interpolating the values of the primal and dual variables stored in Step 3.
- Step 5: The results for the trajectory optimization problem of VTVL reusable rockets are output

For the scenarios designed in Section 4, five finite elements exist for the first and second phases of the flight process as low-density collocation points. In Step 1, the guess of initial value of primal variables at each collocation point is set as the initial state of the reusable rocket, and the guess of initial value of dual variables at each collocation point is set as the default setting in IPOPT. In Step 4, 10 finite elements exist in the first phase, and 15 finite elements exist in the second phase as high-density collocation points. The collocation points of each finite element are three-order Radau points (*i.e.* $K = 3$).

4. Results and discussion

This section presents the results of simulations performed in an AMPL environment (Fourer, Gay, and Kernighan 1990) on a Lenovo Y430p running Windows 7 with an Intel® Core™ i7-4710MQ

Table 1. Simulation parameters.

Parameter	Value	Parameter	Value
m_{10}	$431.6 \times 10^3 \text{ kg}$	$q_{\max}$	80.0 kPa
m_{20}	$107.5 \times 10^3 \text{ kg}$	$acc_{\max}$	$10.0 g_0$
m_{1fuel}	$409.5 \times 10^3 \text{ kg}$	ρ_0	1.225 kg/m^3
m_{2fuel}	$103.5 \times 10^3 \text{ kg}$	h_0	7200.0 m
r_0	$(5605.2, 0, 3043.4) \text{ km}$	A_{ref}	$4\pi \text{ m}^2$
g_0	9.807 m/s^2	C_D	0.5
μ	$3.986012 \times 10^{14} \text{ m}^3/\text{s}^2$	d_{f2}	6593.145 km
R_e	6378.145 km	e_{f2}	0.0076
I_{sp}	340.0 s	i_{f2}	28.5°
$T_{\min}$	360.0 kN	Ω_{f2}	269.8°
$T_{\max}$	934.0 kN	ω_{f2}	130.5°

2.50 GHz processor and 4 GB RAM. The version of the adopted NLP solver IPOPT is 3.8.0. The tolerance of IPOPT is set to 10^{-8} . The simulation parameters are listed in Table 1. Two representative scenarios are designed; they are return to a launch site (RTLS) and return to a drone ship (RTDS). The glide slope angle θ_{gs} for the return phase of the first stage is constrained to 80° . Comparisons are conducted in both scenarios with the conventional initialization strategy, in which the guess of initial value of primal variables at each collocation point is set as the initial state of the reusable rocket, and the guess of initial value of dual variables at each collocation point is set as the default setting in IPOPT, to illustrate the effectiveness of the designed internal–external–growth initialization strategies. The collocation points of the conventional initialization strategy are the same, with the high-density collocation points of the internal–external–growth initialization strategies.

4.1. Return-to-launch-site scenario

In this scenario, the first stage of the rocket returns to the launch site, *i.e.* $r_{f1} = r_0$. For this scenario, the solving process using the finite-element collocation approach with conventional initialization strategy is unsuccessful, whereas the proposed direct trajectory optimization framework can successfully obtain convergent solutions. The time cost for calculating the final high-density discretized solution using the guess of initial value given by interpolating the low-density discretized solution is 1.9 s (CPU time) with 115 iterations. The results of the optimized solutions for the RTLS scenario are shown in Figures 2–4. In the figures, the lines with circle markers represent the results before stage separation of the rocket (the first phase), and the lines with upward-pointing triangle markers represent the results after stage separation (the second phase). Figure 2 shows the time history of the position vector. The position vector (x, y, z) is transformed into the vector of longitude, latitude and altitude. Figure 2(a)–(c) shows the results of the second stage, and Figure 2(d)–(f) the results of the first stage. The time of stage separation is 131.4 s. After stage separation, the second stage continues to deliver the payload to the desired orbit until the time is 453.4 s, and the first stage starts to return to the launch site until the time is 569.7 s. As shown in Figure 2(d)–(f), the first stage still rises after stage separation, goes down after 277.5 s, and lands at the position of the launch site.

The time history of the velocity magnitude of the second and first stages is shown in Figure 3(a) and (d), respectively. From Figures 2 and 3(a) and (d), the first and second stages move together at the same velocity and position before stage separation. In the first phase, the rocket velocity monotonously increases, and the first stage of the rocket is in the full-thrust performance powered by nine engines shown in Figure 3(f). When the thrust is full, the mass of the first stage decreases linearly with the fastest consumption rate shown in Figure 3(e). The engine of the second stage does not work in the first phase; thus, the mass of the second stage remains unchanged, as shown in Figure 3(b). In the second phase, the velocity magnitude of the second stage monotonously increases until delivering the payload to the desired orbit. The single engine of the second stage ignites after stage separation, and the engine is in the full-thrust performance in the second phase shown in Figure 3(c). The mass

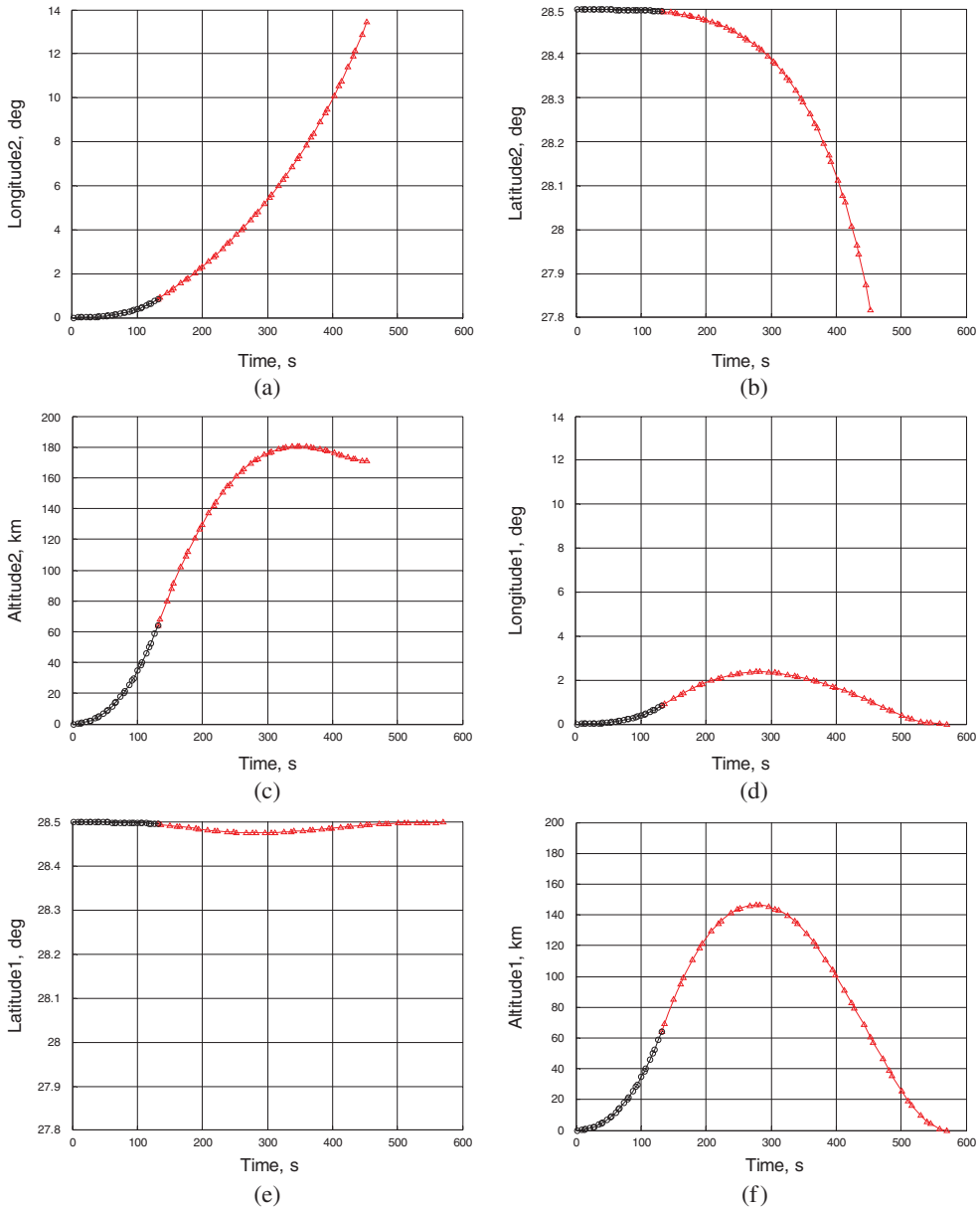


Figure 2. Time history of position vector.

of the second stage decreases linearly at a constant consumption rate, and the final mass of the second stage is $17.31 \times 10^3 \text{ kg}$, as shown in Figure 3(b). After stage separation, the first stage is powered by one engine which is throttleable in the return phase. The velocity magnitude of the first stage remains decreasing for approximately 150.6 s, then remains increasing for approximately 189.7 s, and decreases to zero at touchdown. From Figure 3(f), the thrust magnitude of the first stage in the second phase has a 'bang-bang' profile. When the first stage lands at the launch site, the mass of the first stage is $22.1 \times 10^3 \text{ kg}$, as shown in Figure 3(e). Consequently, the fuel is used up, which is consistent with the performance index to maximize the remaining fuel in the second stage of the rocket.

From Figure 4, the dynamic pressure and visual acceleration of the rocket in the entire flight process satisfy the corresponding constraints. The time history of the dynamic pressure of the second

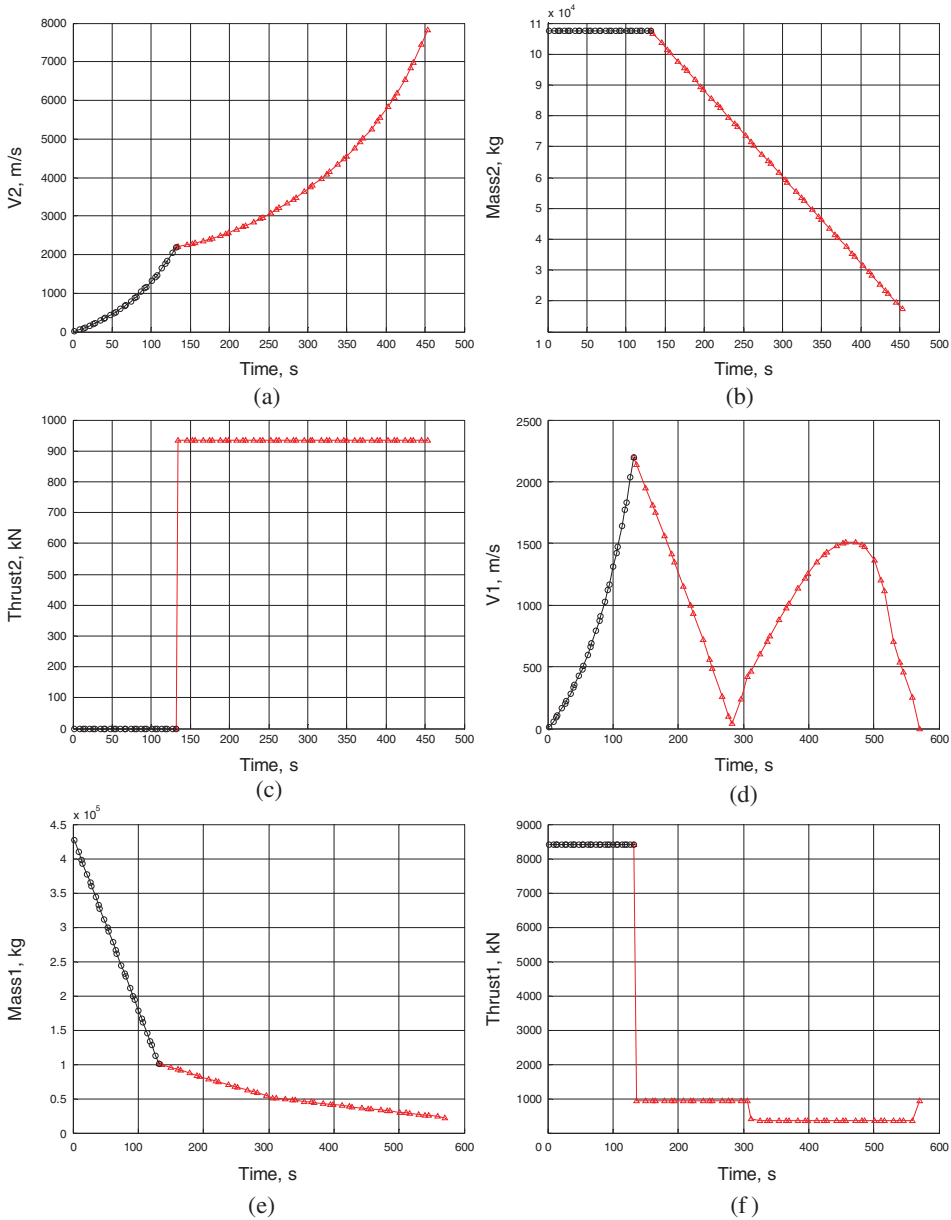


Figure 3. Time history of velocity (V), mass and thrust.

and first stages, is shown in Figure 4(a) and (c), respectively. The dynamic pressure of the rocket first increases after liftoff for approximately 54.6 s and then decreases to zero. After stage separation, the dynamic pressure of the second stage remains zero because the second stage has gone into the vacuum environment. For the first stage, the altitude still rises after stage separation, and the dynamic pressure remains zero. After approximately 470 s, the dynamic pressure of the first stage sharply increases to the upper bound. The first stage then keeps at the bound of the dynamic pressure for approximately 24.7 s, after which the dynamic pressure decreases to zero until the first stage lands at the launch site.

As shown in Figures 2–4, the results for the RTLS scenario are obtained by the proposed direct trajectory optimization framework. As high-density collocation points, 10 and 15 finite elements are

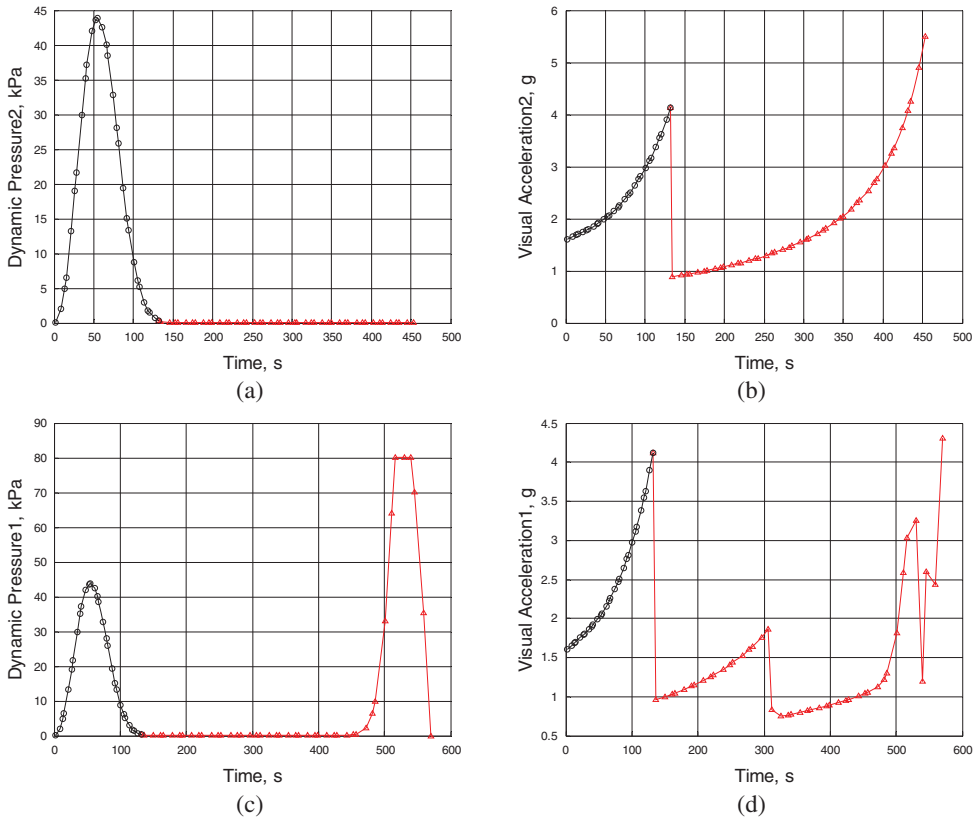


Figure 4. Time history of dynamic pressure and visual acceleration.

present in the first and second phases of the flight process, respectively. Then, the collocation mesh is further refined. The number of finite elements in the first phase varies from 11 to 20, and the number of finite elements in the second phase varies from 16 to 25 accordingly. Therefore, the number of total finite elements varies from 37 to 45 in intervals of two. The guess of initial value is given by interpolating the results of the original high-density discretized solution. IPOPT solves all the resulting NLP problems successfully. The comparison indicates that the discretized solution arising from the original high-density collocation coincides with the solutions arising from the further refined high-density collocation very well, which justifies convergence of the former.

4.2. Return-to-drone-ship scenario

In this scenario, the first stage of the rocket returns to a drone ship. The solving process using the finite-element collocation approach with conventional initialization strategy is also unsuccessful for this scenario, whereas the proposed direct trajectory optimization framework can successfully obtain convergent solutions. The latitude and longitude of the drone ship are 28.0° and 1.0° , respectively. The time cost for calculating the final high-density discretized solution using the guess of initial value given by interpolating the low-density discretized solution is 1.6 s (CPU time) with 90 iterations. The results of the optimized solutions for the RTDS scenario are shown in Figures 5–7. Figure 5 shows the time history of the position vector. Figure 5(a)–(c) shows the results of the second stage, and Figure 5(d)–(f) the results of the first stage. The results indicate that the time of stage separation is 134.6 s. After stage separation, the second stage continues to deliver the payload to the desired orbit until the time is 453.0 s, and the first stage starts to return to land at the drone ship until the time is 528.4 s.

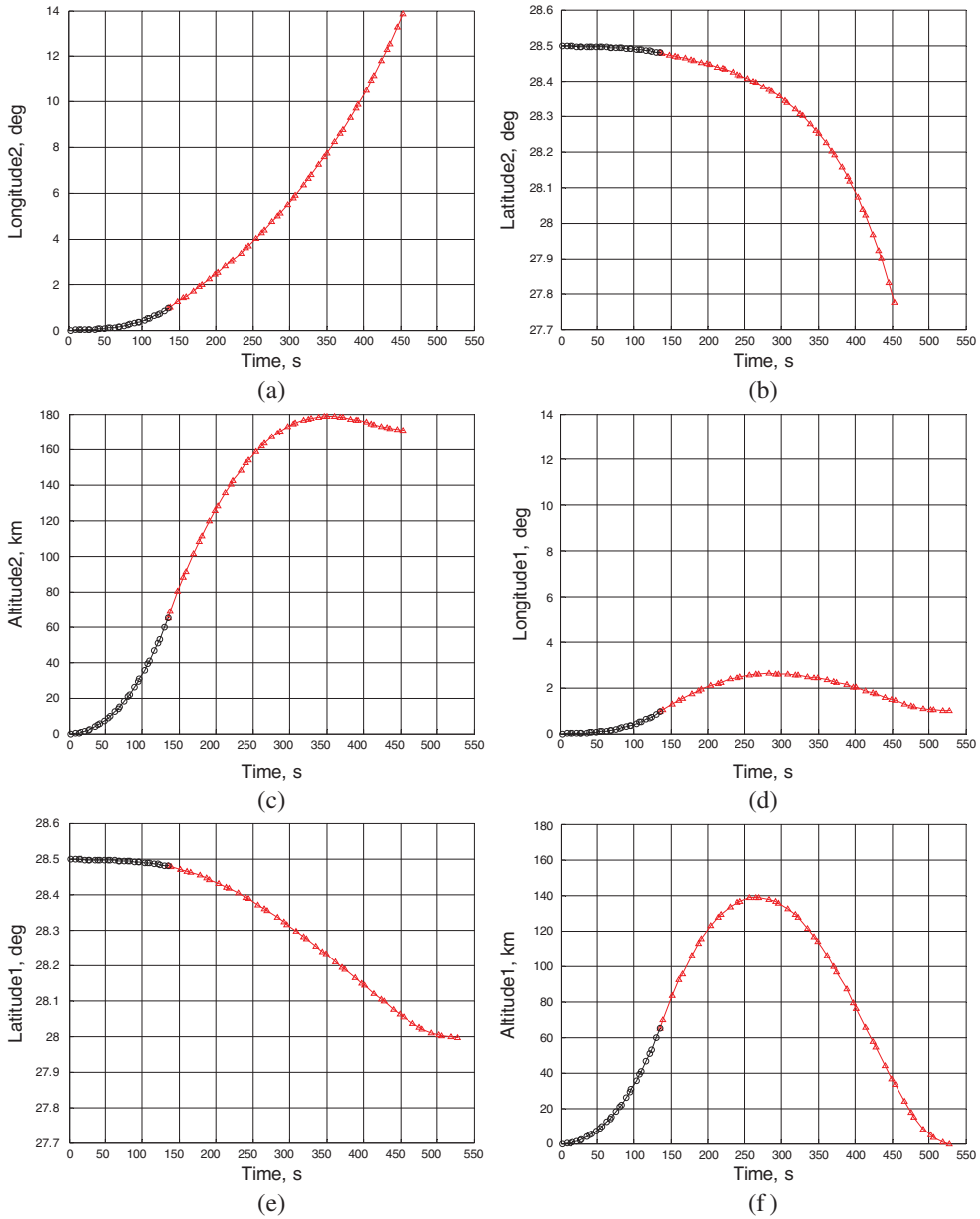


Figure 5. Time history of position vector.

As shown in Figure 5(d)–(f), the first stage still goes up after stage separation, then goes down after 265.8 s, and lands at the drone ship with latitude and longitude of 28.0° and 1.0° , respectively.

The time history of the velocity magnitude of the second and first stages is shown in Figure 6(a) and (d), respectively. In the first phase, the velocity of the rocket monotonously increases, and the first stage of the rocket is in the full-thrust performance powered by nine engines shown in Figure 6(f). As shown in Figure 6(e), the mass of the first stage decreases linearly with the fastest consumption rate when the thrust is full. The engine of the second stage does not work before stage separation; therefore, the mass of the second stage remains unchanged, as shown in Figure 6(b). After stage separation, the velocity magnitude of the second stage monotonously increases until delivering the payload to the

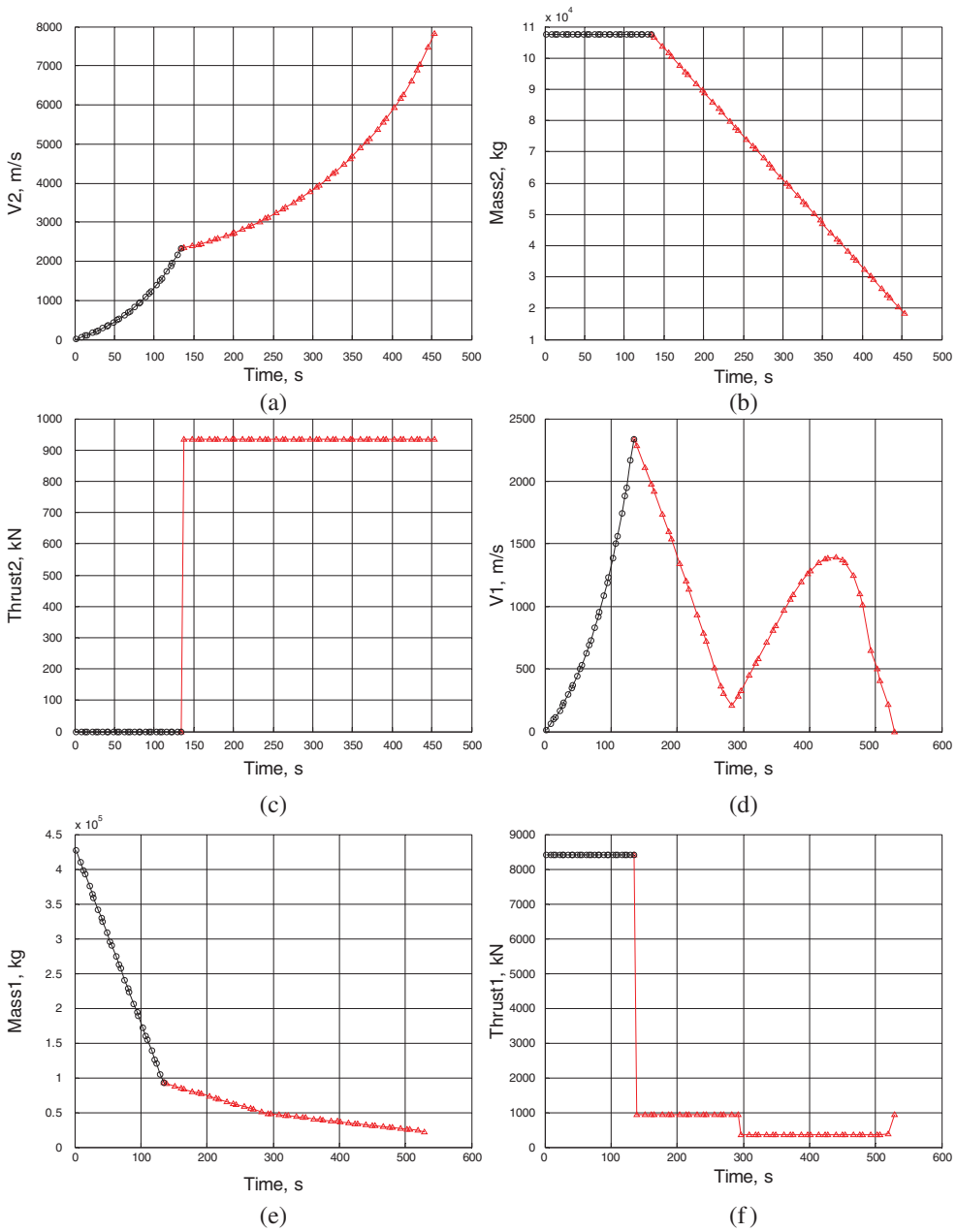


Figure 6. Time history of velocity (V), mass and thrust.

desired orbit. The single engine of the second stage ignites in the second phase, and the engine is in the full-thrust performance shown in Figure 6(c). The mass of the second stage decreases linearly at a constant consumption rate shown in Figure 6(b), and the final mass of the second stage is 18.30×10^3 kg. In the second phase, the first stage of the rocket is powered by one engine which is throttleable. The velocity magnitude of the first stage remains decreasing for approximately 148.2 s, then remains increasing for approximately 157.5 s, and decreases to zero at touchdown. From Figure 6(f), the thrust magnitude of the first stage in the second phase has a ‘bang–bang’ profile. When the first stage lands at the drone ship, the mass of the first stage is 22.1×10^3 kg. Thus, the fuel is used up, which is also consistent with the performance index to maximize the remaining fuel in the second stage of the rocket.

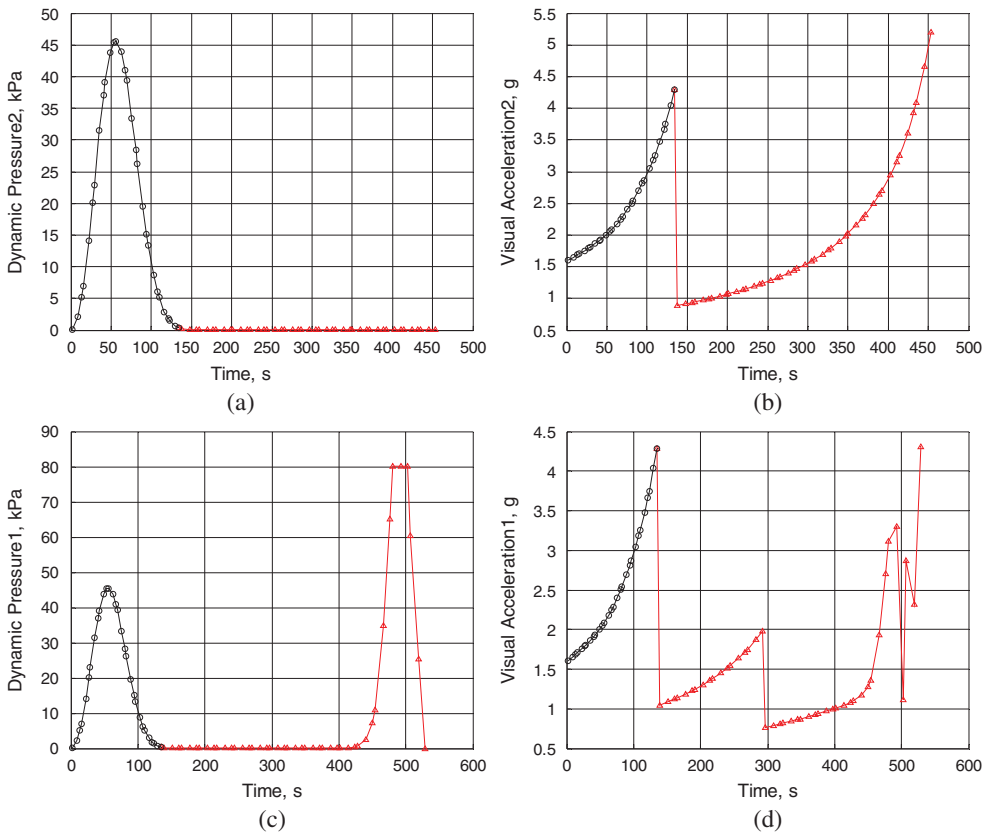


Figure 7. Time history of dynamic pressure and visual acceleration.

As shown in Figure 7, the dynamic pressure and visual acceleration of the rocket in the entire flight process satisfy the corresponding constraints. From Figure 7(a) and (c), the dynamic pressure of the rocket monotonously increases after liftoff for approximately 55.9 s and then decreases to zero. After stage separation, the dynamic pressure of the second stage remains zero because the second stage has gone into the vacuum environment. The altitude of the first stage still increases after stage separation, and the dynamic pressure remains zero. After approximately 427.4 s, the dynamic pressure of the first stage sharply increases to the upper bound. The first stage then keeps at the bound of the dynamic pressure for approximately 22.2 s, after which the dynamic pressure decreases to zero until the first stage lands at the drone ship.

For the designed scenarios, the optimal results obtained by the proposed direct trajectory optimization framework are independently validated by comparing the obtained results with the propagated states via a separate Runge–Kutta propagator. By interpolating the values of the optimal controls at the collocation points and then integrating the dynamic model via MATLAB's ode45 solver, comparisons can be made with the obtained results. The propagated states, which satisfy the boundary and path constraints, are nearly consistent with the optimal results obtained by the proposed direct trajectory optimization framework.

5. Conclusions

A direct trajectory optimization framework for VTVL reusable rockets is presented. The two-phase trajectory optimization problem of a two-stage reusable rocket is established, wherein two stages of the rocket are separately modelled. The established trajectory optimization problem not only ensures

that the first stage of the rocket can return to the specified site, but also guarantees that the second stage can deliver the payload to the desired orbit with maximum remaining fuel in the second stage. The finite-element collocation approach is utilized in the proposed trajectory optimization framework to discretize the original trajectory optimization problem into an NLP problem solved by the NLP solver IPOPT, in which the state and control variables are both discretized. Internal–external-growth strategies are designed to enhance the convergence of solving the problem, in which (1) the problem with simple constraints grows to the problem with complex constraints; (2) a low-density discretized solution grows to a high-density discretized solution; and (3) the values of the primal and dual variables are both considered as the guess of initial value, which considers the character of the primal–dual interior-point method.

The presented direct trajectory optimization framework is applied to two representative scenarios: RTLS and RTDS. The simulation results show that the proposed approach can enhance the convergence of the solving process and exhibits adaptability to deal with missions of VTVL reusable rockets in a systematic way. Although the model of VTVL reusable rockets in this study involves several assumptions for simplification, the proposed direct trajectory optimization framework for VTVL reusable rockets is an open-trajectory optimization framework, and any dynamic, flight-environment or mission-specific constraints can be incorporated in this framework provided that they are explicitly described via equality/inequality.

Disclosure statement

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